Problem 3

In [1]:

```
import numpy as np
```

Training (Constructing Weights)

In [8]:

```
states = np.array([[+1, -1, +1, -1, +1], [-1, +1, +1, -1, +1]])

# Number of states
N = len(states[0])
M = len(states)
# Construct the weight matrix
W = 1 / N * states.T @ states
for i in range(N):
    W[i][i] = 0
W
```

Out[8]:

```
array([[ 0. , -0.4,  0. ,  0. ,  0. ],
       [-0.4,  0. ,  0. ,  0. ,  0. ],
       [ 0. ,  0. ,  0. , -0.4,  0.4],
       [ 0. ,  0. , -0.4,  0. , -0.4],
       [ 0. ,  0. ,  0.4, -0.4,  0. ]])
```

Probing Pattern

In [3]:

```
initial_input_state = np.array([+1, +1, +1, +1, +1])
```

Dynamic Evolution

In [19]:

```
# Hard-limiting non-linearity function
f_h = np.vectorize(lambda x: 1 if x >= 0 else -1)

# Dynamic evolution
def evolve(order, log=False):
    t = 0
    last_U = None
    U = initial_input_state.copy()
    if log:
        print('U({}) = {}'.format(t, U))

    # Until convergence
    while not np.array_equal(U, last_U):
        t += 1
        last_U = U.copy()
        # Update the nodes in order
        for i in order:
            # Be sure to zero-index

            # YOUR CODE HERE
            U[i-1] = f_h(sum(W[j][i-1] * U[j] for j in range(N) if j != i-1))
```

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```
            if log:
                print('U_{}({}) = {}'.format(i, t, U[i-1]))

        # Log U for this iteration
        if log:
            print('U({}) = {}'.format(t, U))
            None

    return U
```

In [21]:

```
U = evolve([3, 1, 5, 2, 4], log=True)
print(U)
```

```
U(0) = [1 1 1 1 1]
U_3(1) = 1
U_1(1) = -1
U_5(1) = 1
U_2(1) = 1
U_4(1) = -1
U(1) = [-1  1  1 -1  1]
U_3(2) = 1
U_1(2) = -1
U_5(2) = 1
U_2(2) = 1
U_4(2) = -1
U(2) = [-1  1  1 -1  1]
[-1  1  1 -1  1]
```

In [22]:

```
U = evolve([2, 4, 3, 5, 1], log=True)
print(U)
```

```
U(0) = [1 1 1 1 1]
U_2(1) = -1
U_4(1) = -1
U_3(1) = 1
U_5(1) = 1
U_1(1) = 1
U(1) = [ 1 -1  1 -1  1]
U_2(2) = -1
U_4(2) = -1
U_3(2) = 1
U_5(2) = 1
U_1(2) = 1
U(2) = [ 1 -1  1 -1  1]
[ 1 -1  1 -1  1]
```

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